Abstract

Technological renewal has played and is still playing a critical role regarding various areas, including professional fields and cultural elements. This thesis aims to bring together two scenarios scarred by digitalization: the work of translator and the popular card game Magic: The Gathering

Its primary objective consists of combining linguistic and translation-related knowledge with basic programming skills in order to develop a prototype of an automatic yet specialized translator using Python. This system is sought to successfully identify and replace the terminology in the game's rules text, which serves as the case study. To achieve it, a system was developed based primarily on the replacement of terms from the source text with equivalent entries identified in a bilingual English-Spanish specialized glossary specifically compiled for this purpose.

Accordingly, the results were analysed through a comparison of the outputs given by each type of translation system. The findings showed that, although general neural machine translation systems offered more fluent and natural linguistic results, they faced difficulties regarding specialized terminology and conventionalized structures. In contrast, the proposed prototype demonstrated greater terminological consistency despite the fact that grammatical or unnatural structures could be encountered, which may be attributed to the lack of machine learning capabilities.

This study highlights not only the importance of technology in contemporary society, but also the indispensable role of the human figure in professional translation processes. Furthermore, it proposes possible improvements to refine the model and enable its potential real-world implementation.

Keywords: Specialized machine translation, computational linguistics, terminological consistency, phraseological applied glossary, board game translation, Magic: The Gathering.

Chapter 6: Conclusions

This dissertation has explored the intersection between linguistics, translation studies and computing through a practical study involving elements that are undoubtedly relevant in today's world. The subject of this study is the collectible card game *Magic: The Gathering* (MTG), or, more specifically, the rules text printed on each card, which is characterised by a high degree of structural constraint and, consequently, lends itself to automation.

Throughout this research, it has become evident that digitalisation and ongoing technological development do not entail the disappearance of human involvement in the translation process. Rather, they require its transformation and evolution towards hybrid, multidisciplinary professional profiles.

To begin with, Chapter 1 provides a historical and contextual analysis demonstrating that adaptability and renewal are inherent characteristics of the translation profession. The evolution of this discipline, from its origins in oral communication to the adoption of Computer-Assisted Translation (CAT) tools and neural machine translation, shows that the profession has endured despite the uncertainty and fear of replacement that have accompanied each technological shift. In the current landscape, the traditional translator profile is evolving and giving rise to a range of specialised roles, such as post-editor, localisation engineer, linguistic data manager, and language model trainer or curator. These roles confirm that the market increasingly demands technical and multidisciplinary skills. This dissertation therefore embraces this emerging professional identity and demonstrates that linguistic skills are not only applicable to technological contexts but also essential to them.

Furthermore, examining the longevity of *Magic: The Gathering*, which could have become obsolete long ago, like many other similar games, has made it possible to understand the game as more than a mere entertainment product: it is also a community in which digitalisation and user-generated contributions have played a crucial role. This makes it an ideal subject for applying linguistic and computational knowledge and developing a prototype programme to assist with its translation, thereby contributing to this demonstrably self-managed ecosystem.

Research into these aspects led to the practical component of the dissertation: the compilation of a bilingual English–Spanish glossary containing terminology and phraseology specific to the game, followed by the development in Python of a specialised machine translation prototype incorporating this glossary. Comparing the official card translations with the output of various general-purpose translation engines (DeepL, Reverso and Google Translate), as well as with the MTG terminology-based system developed for this project, yielded essential insights into the automation of specialised languages.

- **Strengths:**
  The prototype demonstrated superior performance to general-purpose translation engines in terms of maintaining terminological consistency and standardised

collocations. Implementing a hierarchical system that assigned different priority levels to glossary entries greatly helped preserve the fixed structures characteristic of the game's rules text. Furthermore, the exhaustive extraction of terms and an in-depth understanding of the game also proved decisive. Despite the considerable disadvantage of general-purpose engines in this respect, their ability to identify and translate specialised language has shown increasing improvement, probably due to the volume of MTG-related data available online, which these models can draw upon through neural learning.

- **Weaknesses:**
  The prototype exhibited certain shortcomings in grammatical accuracy due to syntactic complexity. In many cases, syntax cannot simply be processed through the substitution of individual components, but must instead be understood as a whole. The engine developed for this project does not understand or translate complete sentences (except for those explicitly included in the glossary). Instead, it operates in a fragmented manner, interpreting the text as a series of isolated "pieces" rather than as a coherent whole. This leads to various errors involving grammar, agreement, clitic placement and meaning. Nevertheless, it should be remembered that this is merely a prototype developed by someone with no prior professional background in technology or programming. Its application in real-world contexts would therefore require thorough review by a qualified professional. By contrast, this ability to produce grammatically fluent output proved to be one of the main strengths of general-purpose machine translation systems.

- **Reflection on the prototype's usefulness:**
  The research confirms that a grammatical error (such as those produced by the specialised prototype) is more noticeable and, consequently, quicker and less costly to identify and correct during human post-editing than a terminological consistency error (typical of general-purpose neural models). The latter may be concealed by apparently natural syntax and therefore go unnoticed by a reviewer without specialised knowledge.

- **Challenges encountered:**
  Several challenges arose during the programme's development, requiring advanced analytical and linguistic skills to resolve. The first involved the repetition of certain terms in the glossary because they appeared within different collocations or structures. This was addressed by introducing a "priority" column, which assigned greater importance to longer linguistic structures. Other problems emerged after the code had been developed, including terminological over-substitution, illustrated by the error in which "graveyard" was translated as "cementrenio" due to interference from the term "enter", and the detection of unintended substrings, as illustrated by the case of "csobretrol", where the system detected and replaced "on" within the word "control". Although the documentation required to resolve these errors was computational in nature, identifying and understanding them required substantial translation expertise.

> This leads to the conclusion that programming problems arising in the context of translation are, ultimately, problems of a philological and textual nature.

Although the results were satisfactory in terms of lexical fidelity and compliance with MTG's established terminology conventions, it is essential to acknowledge the prototype nature of the software developed. In light of this, several potential avenues for further technical development can be identified. First and foremost, priority should be given to adding a third stage to the translation workflow, incorporating a Natural Language Processing (NLP) component. This would make it possible to modify or restructure the resulting syntax after glossary application without altering the terminology that had already been fixed. The result would be a machine translation engine combining the lexical specialisation of the prototype developed for this project with the grammatical accuracy of general-purpose neural translation systems. However, while this approach could produce more natural and fluent syntax, preserving MTG's fixed structures would also require expanding the glossary to include an even wider range of phraseological elements.

Further glossary expansion is also envisaged to include character names, deck names, recurring place names and titles of collections within the Magic universe. This would provide a support tool for human translators working on flavour text, which frequently contains intertextual references to other cards in the game.

This tool was primarily conceived as an aid for professionals responsible for translating cards. However, it could also prove useful in other contexts, such as the development of an application for players providing quick and accessible translations of cards, since many players use decks in English. Furthermore, the structure and methodology of this study could potentially be adapted to other highly formalised language domains governed by rigid structures, such as patents, instruction manuals and standardised legal texts.

Although this study may appear to focus heavily on computational logic, problem-solving skills in translation have in fact played a decisive role throughout the process. Human professionals are always needed to apply their linguistic expertise, whether they are reviewing, correcting, supervising or even developing automation systems. As a final consideration, this dissertation therefore confirms that technology should not be regarded as a threat to the role of translators, but rather as an opportunity to expand their scope and capabilities. The development of this functional prototype is intended not only as a modest contribution to the self-managed ecosystem of *Magic: The Gathering*, but also as empirical evidence that linguistic and translation skills are both applicable and necessary in the development of technological tools.